# Mariamma Antony

PhD Candidate, Department of Computer Science and Automation
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## Research Summary

PhD candidate in the Machine Learning Lab at IISc Bangalore working at the intersection of Machine Learning, Medical AI, Vision-Language Models (VLMs) and Explainable AI. My research focuses on developing robust and interpretable multimodal AI systems for medical imaging, radiology report generation, model editing, and mechanistic interpretability.

## Research Interests

Vision-Language Models (VLMs), Medical Image Analysis, Explainable AI, Large Language Models, Model Editing, Task Arithmetic, Mechanistic Interpretability, Multimodal Learning, AI for Healthcare.

## Education

**Indian Institute of Science (IISc), Bangalore** 2019–Present
PhD Candidate, Department of Computer Science and Automation
Advisor: Prof. Chiranjib Bhattacharyya

**Indian Institute of Technology (IIT) Kharagpur** 2013
M.Tech., Computer Science and Engineering
CGPA: 9.12/10

## Publications and Preprints

**CheXWhatsApp: A Dataset for Exploring Challenges in the Diagnosis of Chest X-rays**
Mariamma Antony, Rajiv Porana, S.M. Lathiya, S.T. Kakileti, Chiranjib Bhattacharyya
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025*

**Challenges of AI-driven Diagnosis of Chest X-rays Transmitted Through Smartphones: A Case Study in COVID-19**
Mariamma Antony, S.T. Kakileti, R. Shah, S. Sahoo, Chiranjib Bhattacharyya, G. Manjunath
*Scientific Reports (Nature Portfolio), 2023*

**TACS-Guided Self-Alignment of LVLMs for Explainable Chest X-ray Analysis**
Mariamma Antony, Rajiv Porana, Prathigya Ranjit, Chiranjib Bhattacharyya
*OpenReview, 2025*

**Learning High-Fidelity Rulesets using Submodular Maximization**
Mariamma Antony, Raman Sankaran, Chiranjib Bhattacharyya, Uma Satya Ranjan
*arXiv, 2025*

**Finding a Small Set of High-Degree Nodes in Time-Varying Graphs**
Mariamma Antony, Arobinda Gupta
*IEEE WoWMoM, 2014*

## Manuscripts

**Task Arithmetic for Model Editing**
Mariamma Antony, Sam Betson, Chaitanya Murti, Chiranjib Bhattacharyya
- Investigating failure modes of existing task arithmetic approaches.
- Developing optimization-based formulations for model composition and editing.
- Received three Weak Accept and one Weak Reject reviews at ICML; revised manuscript incorporating reviewer feedback is currently under review at NeurIPS.

**Multi-Task Learning of Vision-Language Models for Tuberculosis Diagnosis using Chest X-ray Images**
Mariamma Antony, Sam Betson, Rajiv Porana, Chiranjib Bhattacharyya
- Developing robust VLMs for tuberculosis diagnosis from chest X-rays by collaborating with radiologists from Father Muller Medical College.
- Manuscript completed and prepared for submission to a leading machine learning conference.

## Ongoing Research

**Vision-Language Models for Bone Metastasis Report Generation**
Collaborators: AIIMS Bhubaneswar
- Developing domain-specific VLMs for automated report generation from bone scan images.
- Training vision encoders and language models using paired image-report datasets.

**Mechanistic Interpretability of Vision-Language Models**
- Investigating attention heads and multimodal reasoning mechanisms in LLaVA-style architectures.
- Developing attribution methods for explainable multimodal AI systems.

## Research Leadership and Mentoring

- Led interdisciplinary collaborations involving IISc, AIIMS Bhubaneswar, and Father Muller Medical College.
- Mentored 9+ M.Tech, undergraduate, and industry interns on projects in Medical AI, VLMs, and Explainable AI.
- Coordinated development of large-scale medical imaging datasets and annotation pipelines.

## Research Collaborations

- Father Muller Medical College, Mangalore
- AIIMS Bhubaneswar
- Machine Learning Lab, IISc Bangalore
- Clinical collaborators in Radiology and Nuclear Medicine

## Technical Skills

**Programming:** Python, Java, SQL, JavaScript

**Machine Learning:** PyTorch, HuggingFace Transformers, Deep Learning, LLMs, Vision-Language Models

**Research Areas:** Medical AI, Explainable AI, Trustworthy AI, Model Editing, Mechanistic Interpretability

**Distributed Systems:** Apache Spark, Kafka, Storm

## Industry Experience

**American Express, Bangalore** 2015–2019

Engineer II → Engineer I

- Developed large-scale machine learning and distributed systems using Spark, Kafka, and Storm.
- Led development of forex prediction systems for cross-border payments.
- Built enterprise-scale streaming systems for financial transaction processing.

**Yahoo, Bangalore** 2013–2014

Senior Software Engineer

- Developed APIs supporting Yahoo News, Finance, Sports, and Video platforms.
- Improved scalability and throughput of media services.

## Awards and Achievements

- All India Rank 350 in GATE 2011 among 136,027 candidates.
- Ranked 15 among 60,000+ participants in Geek Goddess 2018.
- American Express Global Commercial Services Ambassador Award (2018).
- American Express Best Industry Contributor Award (2017).
- Yahoo Spot Award for Performance Improvement (2015).
- Founded Women in Technology (WIT) Chapter at American Express Bangalore.